CoRaw (logo)

CoRaw is a webapp designed to help users learn and practice conversational Irish using AI and meta-prompting.

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Interim Report – CoRaw

1.0 INTRODUCTION

My name is Oisín Larkin, and I am completing a Higher Diploma in Computer Science at SETU. My project that I am doing in order to complete the higher diploma is a Web App called CoRaw. The idea behind this web app is that it will help provide anyone learning the Irish language with a platform to practice speaking the language and develop their Oral skills. The reason I landed on this idea is that I am someone who is trying to learn the language as an adult and in from my experience I felt that this is something that would be very useful to me. There is a lot of material to read and listen to, but the opportunity to actually practice speaking and using the language is very rare and can also be stressful. I hope my app can provide people with a stress-free place to speak and use the language where they will be aided in vocabulary expansion in important areas of conversation.

CoRaw is a webapp designed to help users learn and practice conversational Irish using AI and meta-prompting. The webapp will take the user through immersive realistic conversations that will translate to improving real world conversational skills in Irish.

The idea is that CoRaw will give people who are learning the Irish language a chance to engage in realistic everyday conversations at their own pace with an AI model.

Users will be presented with some questions when they first login to determine their level and topics they are interested in. CoRaw will then present them with their own learning dashboard. This dashboard will have their suggested conversations and will also have transcripts they can access of their previous chats to be used as a learning resource.

Each conversation will also have its own accompanying vocabulary resource where the user can get some basic words and phrases to get them ready should they need it.

The chats/roleplays that users can engage in will be based on meta prompting whereby when they enter the conversation, both the user and model are aware of the conversations pre coded context. The idea is that this will enable the user to have a somewhat structured and realistic environment where they can practice conversing about different topics and build their confidence in speaking the language. This removes the barrier of having to find a speaking partner and also removes the pressure to speak fluently straight away.

The app will allow the user to translate sentences during the conversation, or to have sentences repeated to them slowly. The idea here is that the app is useful and accessible to people of all levels and that it maximises the opportunity to not only learn new phrases or words but to practice speaking with them straight away.

1.1 Motivation

I am doing this as I have recently found a passion for learning Irish, but I find it hard because there is not much opportunity to actually speak the language. Speaking Irish with AI without any context or goals seems to be an endless loop. So, hopefully this app can tackle that problem by offering a structured path of progress to having a stronger conversational ability. The useful and likely conversations one might have will be all listed out and ready to let you start speaking straight away and the app will ensure that conversations stay on a path that will provide the best learning opportunities. I also find that a lot of people are at an awkward stage with the Irish language where they can converse but not at a level to hold a conversation with a fluent speaker. I think by allowing the user to have translations mid conversation and have statements repeated as much as they want, this web app will help them develop their speaking ability and confidence.

2.0 Specification and Requirements

2.1 System Overview

CoRaw is a web-based application that will be designed to support and enable the development of conversational Irish language skills through structured AI assisted roleplay conversational scenarios.

Users will be enabled to select between a range of conversations based on topic and difficulty depending on their level. During a roleplay the users provide vocal input in response to questions from the system, this will be converted into text using speech recognition technology. The system will then respond with a relevant context-based response using an AI language model before proceeding with the next question for the user.

Following a roleplay, learning will be enhanced with additional learning tools. The system will provide the user with a transcript of each conversation and provide a grammatical review to the user. CoRaw is a web app that will be designed to integrate external AI and other speech technologies to simulate real world conversational practice.

2.2 Functional Requirements

The functional requirements will define the core capabilities that CoRaw will need in order to provide structured conversational Irish language practice to users.

A. **Account Management**

FR1 – The system shall allow the user to create an account

FR2 – The system shall allow the user to sign in to their account

FR3 – The system shall allow the user to select their level

B. Conversation Management

FR4 – The system shall allow the user to select a predefined conversational scenario

FR5 – The system shall initiate the conversations with predefined contextual prompts

FR6 – The system shall accept spoken input from the user

FR7 - The system shall convert speech input from the user into text

FR 8 – The system shall generate a contextual response for the user using an AI language model

FR9 – The system shall provide the response to the user in both text and speech format

C. Learning Support Features

FR10 – The system shall provide relevant vocabulary assistance to the user for each conversation

FR11 – The system shall provide a transcript after every completed conversation

FR12 - The system shall provide a translation functionality upon user requests

FR13 – The system shall provide grammar reports on the user's transcripts.

2.2 Non-Functional Requirements

A. Performance

NFR1 – The system will respond to the user with an acceptable delay

B. Usability

NFR2 – The system shall have an intuitive user interface design

NFR3 – The system shall provide clear and effective speech recognition

C. Reliability

NFR4 – The system shall handle external API and database errors gracefully

D. Security

NFR5 – API keys and other sensitive information shall not be exposed in code

NFR6 – The system shall not store sensitive personal data

3.0 BACKGROUND RESEARCH AND LITERATURE REVIEW

3.1 Irish Language Learning Context

3.1.1 Target Users

As stated in the introduction. The target users of this app are people who have some understanding of the Irish language (studied it in school for example) who would like the opportunity to practice speaking it more. This could be someone who is returning to learning the language after not speaking it in years. It could be a student in the middle of their studies. It could be someone who has a passion for speaking the language but does not have anyone to speak it with. Similarly, it could be someone who is perhaps not yet comfortable to start speaking the language.

3.1.1 User Goals

The main goals that a user of this web app may have is all to do with developing their oral skills with the language. Without having a personal tutor or a fluent speaker, opportunities to practice speaking a language can be rare. This is particularly true with the Irish language as it is a minority language. Fluent speakers are limited and personal tutoring could be expensive. Outside of small pockets of the country real life scenarios where someone who is not fluent could get oral experience with the language are rare. The app aims to help these users achieve their goal of improving spoken ability, fluency and confidence by providing them with their own environment for spoken practice that they can use at any time.

3.2 Industry Analysis

When looking at the market of language learning web apps or even mobile applications, there are a lot of different tools and options.

3.2.1 Gamified Language Platforms

The biggest and most popular online language learning resource right now is a gamified learning app called Duolingo. This app brings the learner through a series of gamified lessons in order to expand their vocabulary.

3.2.2 Social / Peer Learning Platforms

Other apps such as HelloTalk are more of a social platform designed to help users connect with native speakers with goal of learning the language that way.

3.2.3 Tutor-Based Platforms

Other language learning apps connect users with tutors online who can then give them lessons in languages of their choice.

3.2.4 AI-Based Conversational Platforms

The closest competitor to CoRaw is an app called Langua. It offers similar conversations or roleplays with predefined context on different scenarios and topics for learning. However, like many apps, while it offers support for a range of different languages, this does not include Irish.

3.2.5 Identified Market Gap

There are a few different problems that while these apps are good learning tools, I hope that CoRaw will address that these apps do not. While a lot of apps are based around expanding vocabulary, they do not offer as much opportunity for the person to really use the language without a lot of guard rails in place maybe perhaps just saying very specific words or sentences. With the learning environment CoRaw hopes to create, the learner should learn how to use the vocabulary that they have and not just expand it. They should be able to practice engaging with the language in a purely practical way, improving their ability and confidence to actually use the language.

While with a personal tutor you can get this experience of learning from a pure immersion and oral approach, this can be extremely costly. While apps like preply offer the chance to engage in this type of learning with a native speaker, having to pay per lesson and for a private tutor will add up in a way that using an app like CoRaw that uses ai speech to text and text to speech wont.

CoRaw also addresses the issue of being aimed at learners of the Irish language. The Irish language being such a small and minority language, finding full and expansive support on different apps can be difficult.

3.3 Open-Source and Repository Analysis

3.3.1 Abair

Abair is a leader in speech and language AI technologies in the Irish language. The initiative is from the Phonetics and Speech Laboratory, School of Linguistic, Speech and Communication Sciences of Trinity College and is funded by the Department of Rural and Community Development and the Gaeltacht. Abair is developing AI technologies for the different dialects of Irish and has created different resources and applications with these technologies.

Its main technologies write now are Abair which was their first technology which is an effective text to speech technology for the Irish language. They also have Éist an effective speech to text technology for Irish. The are also currently working on a Comhrá (conversation) section on their webapp, allowing you to speak with an AI in the different dialects of the Irish language. They also contribute to a range of apps aimed at education and accessibility as well as public apps that allow users to utilise their technologies.

3.3.2 Gramadóir

Gramadóir is an open source grammar checking engine. It was developed as a tool to aid the development of natural language processing tools. It was purposely designed for the Irish language and since ports for other languages such as Afrikans, Akan Cornish, Scottish Gaelic and more have been attempted. In 2021 a coud based grammar checking version was launched and in 2018 a public API was released, allowing developers to incorporate the checker into their own apps.

3.3.3 An-Scealaí

An scéalaí is an online Irish language learning platform for students. It provides users with an environment where they can write and correct text in Irish. The two main features of An-Scéalaí is that it allows users to hear their stories spoken back to them by integrating Abair, as

well as this An-Scéalaí allows users to receive grammar feedback as it also incorporates An Gramadóir.

3.4 Academic Literature Review

3.4.1 Speech-Based Learning and Oral Proficiency Development

Research on second language acquisition consistently emphasizes the importance of speaking a language in order to develop overall competency and proficiency. Oral practice allows learners to effectively use their language and receive real time feedback, whether explicit (direct correction) or implicit (misunderstood). This active and interactive learning has been shown to improve fluency, confidence and accuracy in speech (Long, 1996).

Swain's output hypothesis emphasises this. It claims that pushes learners from a purely receptive form of learning to speaking a language enables them to develop their ability to express themselves with the target language. Empirical studies support this hypothesis that regular meaningful conversation develop superior language skills when compared to their counterparts who focus predominantly on receptive skills such as reading and listening (Swain, 2005).

3.4.2 AI and Language Learning

When discussing advantages of using AI tools for language learning the literature explores both. In terms of advantages there are lots of tools, especially more recently that help language learners to save time and speed up their learning. Some of these tools provide real time translations, language tutoring systems or even personalised lessons and feedback. In the last few years, they have become especially accessible and increasingly utilised. Common examples are Duolingo, Elsa Speak and even Google translate (De le Vall et al, 2023).

Despite the accessibility and effectiveness that AI provides, the literature also touches on some of the downsides to it. Learning a language through real life interaction and as part of a classroom community could lead to enhanced learning. It can also provide the opportunity for students to utilise more realistic and authentic language. Group scenarios may also help learners to develop better socio-pragmatic awareness in their language. Despite the rapid improvement in AI language tools, it is still difficult to replicate the learning environment of a real-world conversation (Kannan and Munday, 2018).

4.0 TECH FEASIBILITY

4.1 Frontend Technology Selection (React + Typescript)

React is a frontend framework and JavaScript library for web and native user interfaces. React allows users to build user interfaces out of multiple individual pieces the user can create called components. The components can be combined to create screens pages and apps. React components are also good for dynamic web apps as they can receive data and subsequently return what should appear on screen. Since CoRaw will be particularly dynamic with loading responses from the system and loading in user input the React framework is a suitable choice. React also uses a virtual DOM mechanism that enables it to make only the necessary updates to the real DOM making it fast and efficient.

To compliment the use of React for CoRaw I also decided to use Typescript. This is due to the advantages it provides in terms of code reliability, maintainability and scalability. The use of explicitly defined data structures and interfaces pairs well with react components and enforces type safety at compile time. Typescript reduces likelihood of runtime errors and improves code standard which is useful especially when handling a lot of data transfers and conversations which CoRaw will have to do.

4.2 AI Integration (OpenAI API)

I have decided to use OpenAI API for providing the system conversation responses. The reason I have decided to do this is that it can be called as an API which is less resource intensive than hosting a model. Due to the large dataset that OpenAI models are trained on and their proficiency with minority languages such as Irish it is an appropriate option for generating contextual responses to CoRaw users.

4.3 Speech Recognition

For speech recognition and text to speech handling I needed to find an API that can do Irish language speech to text. While Abair have their own tool that has effective speech to text, there is no public API for it. However, I did find a public API that has effective Irish language speech to text called Speechmatics which I will user for CoRaw to take in User input and convert it to text.

4.4 Backend Framework Architecture

I will use node.js for the backend for its modularity and large ecosystem of libraries and tools. Node.js is lightweight and ideal for external API calls which will be a large feature in CoRaw. Node.js will be paired with the hapi framework also due to its modular architecture and built in validation systems as well as its other useful plugins that CoRaw will implement.

4.5 API Design

The CoRaw backend architecture will be implemented as a RESTful API. This will ensure that there is a clear separation between the frontend and the backend. The frontend will communicate with the backend through well defined and documented Http endpoints. Swagger (OpenAPI) will be integrated to document and define the API specification.

4.6 Database Selection – MongoDB

CoRaw will use MongoDB. I decided to use MongoDB due to its flexibility and document orientated data model. This suitable for the dynamic nested data structures that CoRaw conversations will generate. Using a NoSQL database is also a flexible option that will provide adaptability during the development process of the webapp. The scalability and ease of deployment will also be useful for development.

5.0 METHODOLOGY AND PLANNING

5.1 Chosen Development Methodology

For CoRaw I decided to use Agile development methodology for multiple different reasons. Agile methodology is easily and quickly adaptable. As I am working full time while building CoRaw, my plans and sprints can quickly change and with I can add these changes to my backlog easily. I also like the sprint structure of Agile and having small goals to work toward that are broken down, rather than bigger longer-term goals. I also found that with agile having a check in with myself / standup everyday kept me aware of where I am with the project and how it was going which helped prevent plans from slipping away.

5.2 Agile Sprint Structure

The implementation was organised into short sprints. Each spring typically lasted 2 weeks. This enabled the structure to be broken down into small manageable tasks over the course of each sprint. Sprint structure was focused on functionality and started with the core structure. The next sprints then moved into the conversational an additional functionality.

Sprint progress was tracked using burndown charts. After the weekly meeting with my supervisor, if it was the end of the sprint I would use the minutes from the meeting to create a burndown chart. Burndowns charts were useful as they were effective at planning and tracking tasks but also with creating a visual and measurable way of tracking progress.

Figure 1. Burndown Chart 1st Sprint

Sprint Burndown Chart																
Backlog ID	User Stories	Initial Estimate	Feb-06	Feb-07	Feb-08	Feb-09	Feb-10	Feb-11	Feb-12	Feb-13	Feb-14	Feb-15	Feb-16	Feb-17	Feb-18	Feb-19
		Day 0	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9	Day 10	Day 11	Day 12	Day 13	Day 14
	Set up Mongo	3	2													
	Add User Accounts	3	1													
	Login + Onboard UI	6														
	Format + Structure Report	3														
	Create Dashboard UI	10														
	Finish Interim Report	7														
	Feature 8															
	Feature 9															
	Feature 10															
Remaining Effort		32	29	29	29	29	29	29	29	29	29	29	29	29	29	29
Ideal Trend		32	30	27	25	23	21	18	16	14	11	9	7	5	2	0

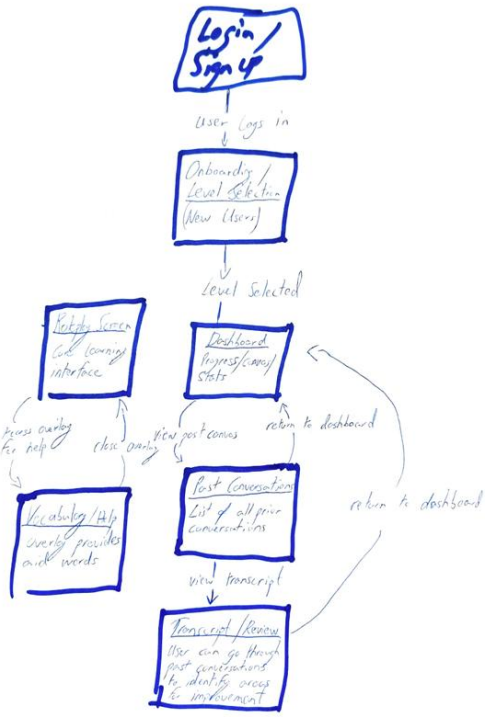
6.0 SYSTEM ARCHITECTURE (To do)

7.0 INITIAL DESIGN AND MODELLING

7.1 System Design Overview (To do)

7.2 User flow diagram

Figure 2. User flow diagram



The user flow diagram shows the general flow of how the user will interact with CoRaw and how the different pages in the app interact with each other. The diagram shows that the user logs in, goes to the onboarding screen to select live, then they are brought to their dashboard. From there the user can go to the roleplay/new conversation screen which has an optional vocabulary overlay. From the dashboard the user can also go to a past conversations page where they can select to view the transcripts of past conversations.

7.3 Wireframes and Interface Design

I used a wireframing tool called Ballsamiq to generate examples of the pages that CoRaw will have.

Figure 3. Login/Signup Page

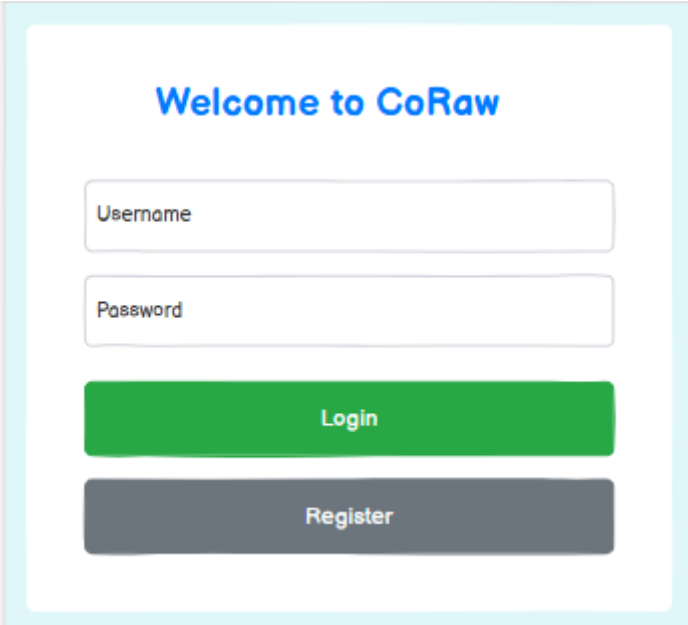


Figure 4. Onboarding Screen

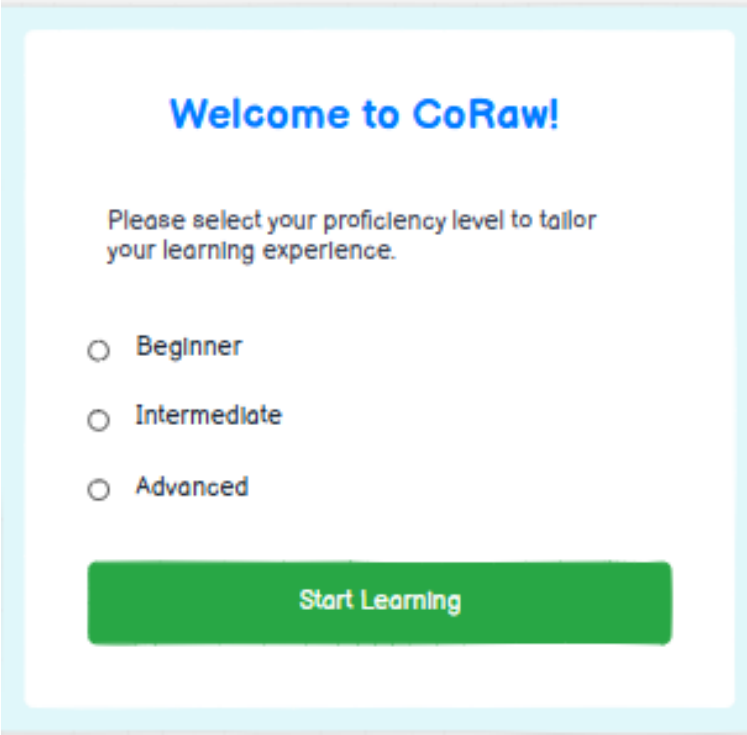


Figure 5. Dashboard

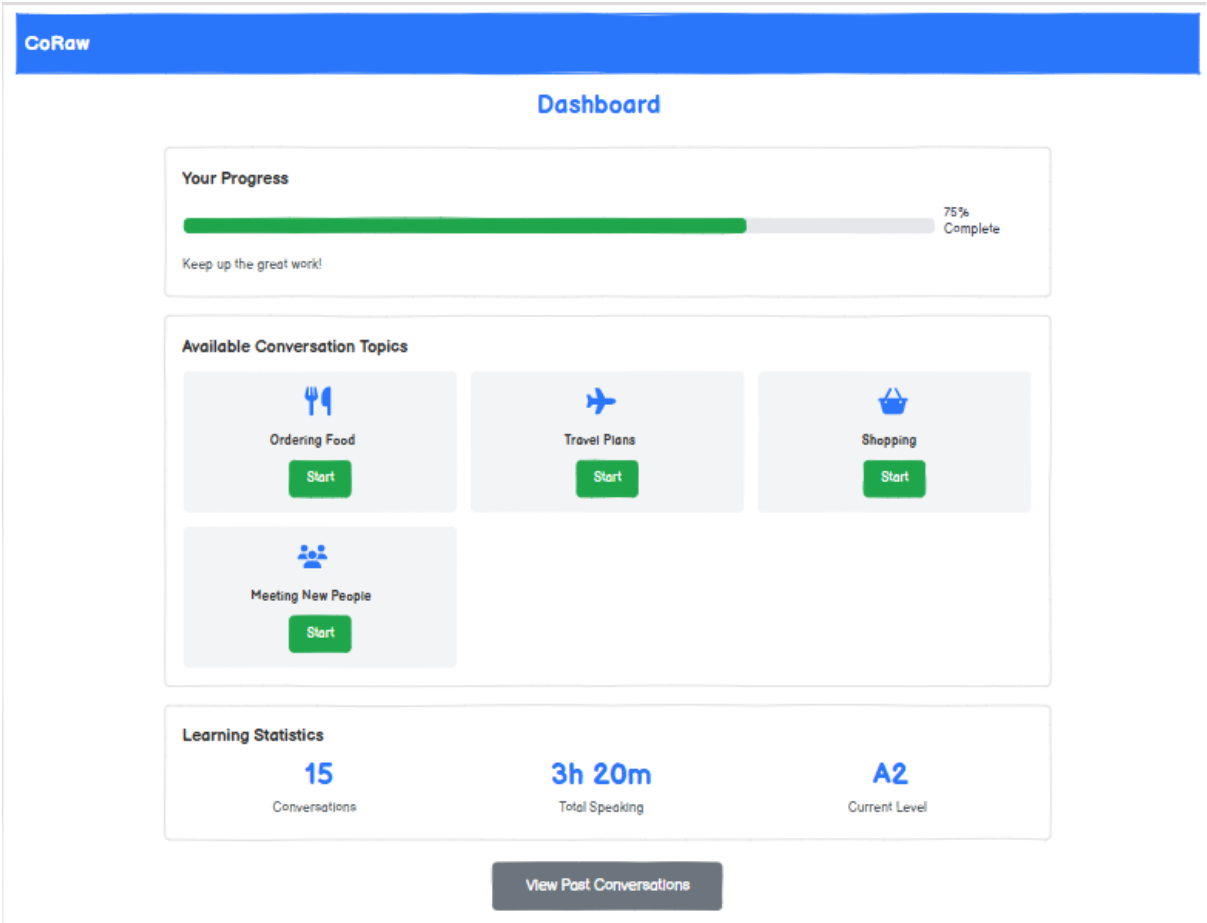


Figure 6. Conversation Screen

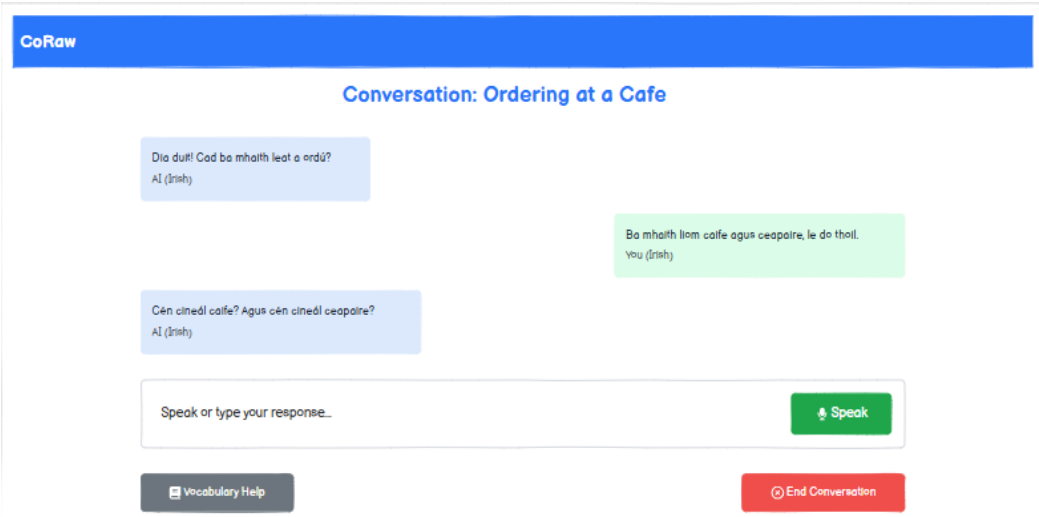


Figure 7. Vocabulary help Overlay

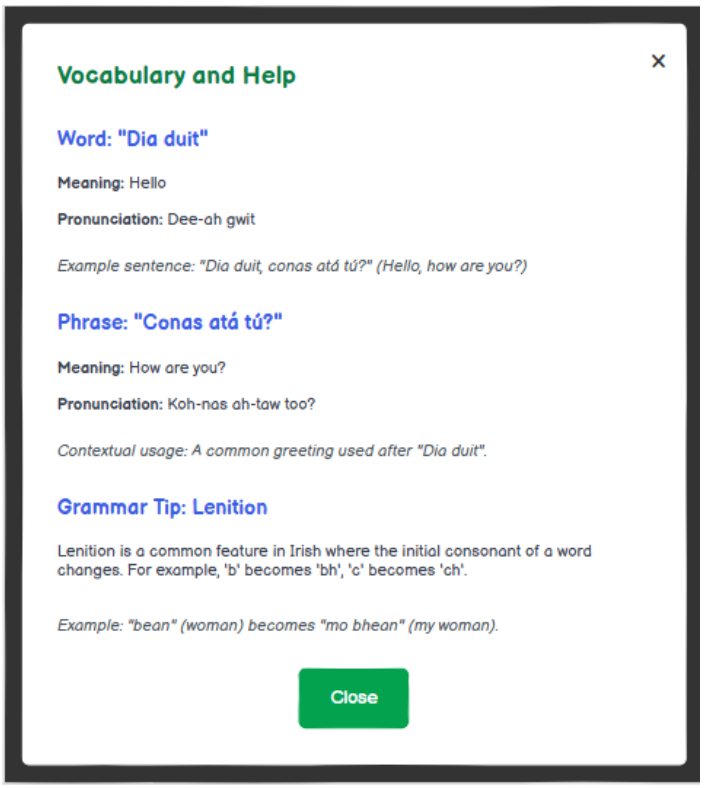


Figure 8. Past Conversations

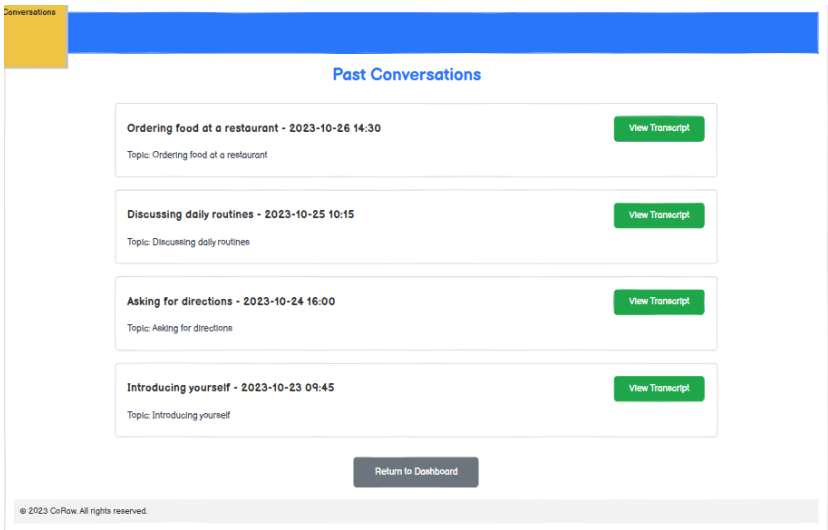
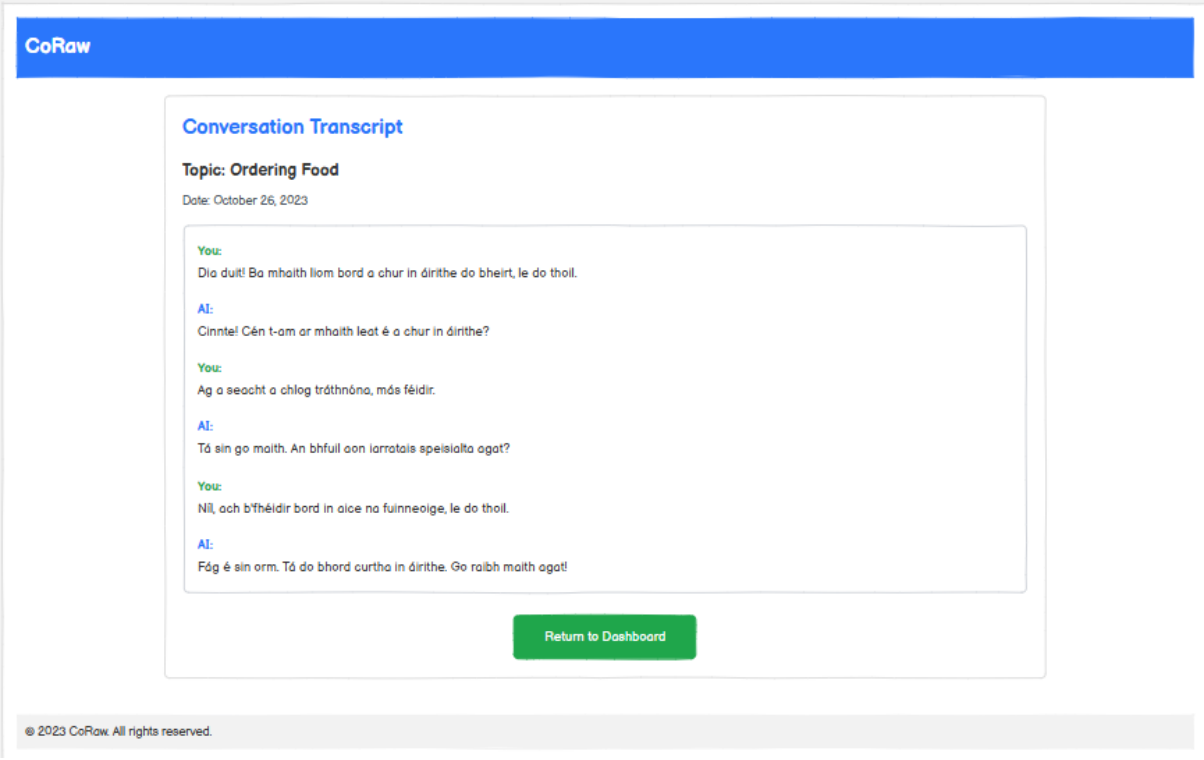


Figure 9. Transcript Screen



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